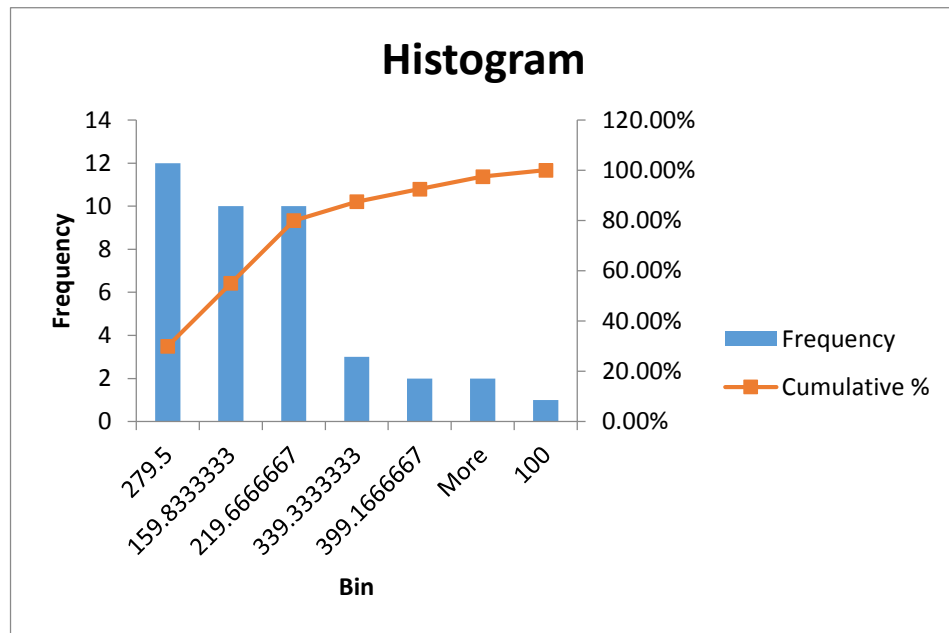


1. Using the following data, construct a histogram of sandwich size in grams

Size in grams
100
279
147
263
269
459
164
273
114
214
209
246
294
147
189
244
165
206
226
249
352
149
190
284
169
220
229
118
376
121
250
190
198
143
260
125
434
133
311
122



<i>Bin</i>	<i>Frequency</i>	<i>Cumulative %</i>	<i>Bin</i>	<i>Frequency</i>	<i>Cumulative %</i>
100	1	2.50%	279.5	12	30.00%
159.8333333	10	27.50%	159.8333333	10	55.00%
219.6666667	10	52.50%	219.6666667	10	80.00%
279.5	12	82.50%	339.3333333	3	87.50%
339.3333333	3	90.00%	399.1666667	2	92.50%
399.1666667	2	95.00%	More	2	97.50%
More	2	100.00%	100	1	100.00%

2. Average heights in a kindergarten class were 40" with a standard deviation of 2". All students with heights from 38" to 46" are considered to be within the national average.

a. If the heights are approximately normally distributed and 14 students are within the national average range, how many students in the class are shorter or taller than the national average?

None of the students are taller than the national average, as the average is 40.

$$(1/2)(14) = 7 \text{ students}$$

b. What is the probability that a student is taller than 44"?

The probability that a student is taller than 44 is $(44 - 40)/2 = 2 \rightarrow$ probability is 95.4%
95.4%

c. What is the probability that a student is shorter than 37"?

The probability that a student is shorter than 37 is $(37-40)/2 = -1.5 \rightarrow$ probability is less than 90%
Less than 90%

Mean 40

Std Dev 2

National Average 38-46

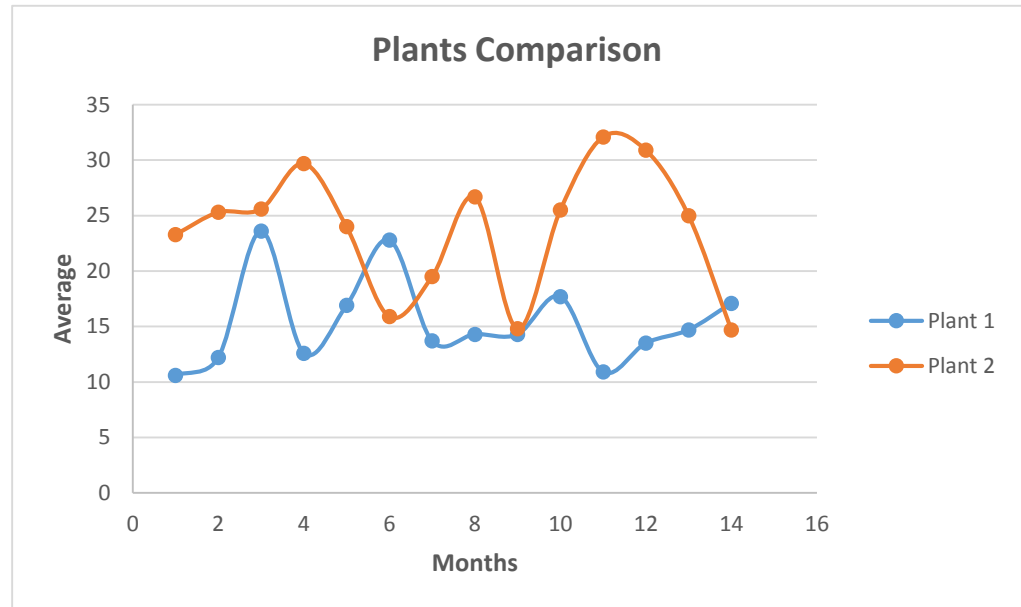
Heights are approximately normally distributed

14 students are within the 38-46 national average range

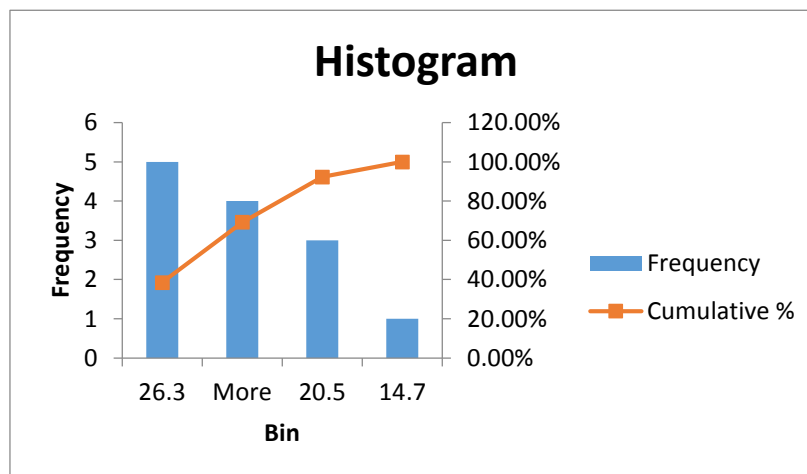
3. The following are the average sulfur dioxide emissions from two different power plants in the Finger Lakes NY area (in average tons per day) for 14 months. For example, Plant 1 in month 1 averaged 10.6 tons per day.

Analyze the data and generate an appropriate graphical representation.

Plant 1	Plant 2
10.6	23.3
12.2	25.3
23.6	25.6
12.6	29.7
16.9	24
22.8	15.9
13.7	19.5
14.3	26.7
14.3	14.8
17.7	25.5
10.9	32.1
13.5	30.9
14.7	25
17.1	14.7

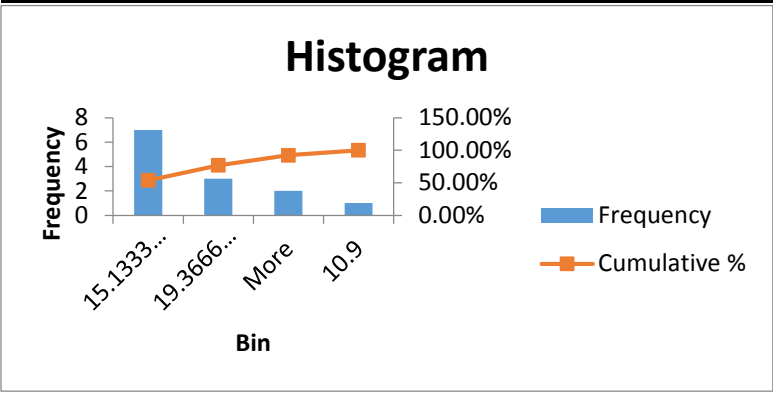


Bin	Frequency	Cumulative %	Bin	Frequency	Cumulative %
14.7	1	7.69%	26.3	5	38.46%
20.5	3	30.77%	More	4	69.23%
26.3	5	69.23%	20.5	3	92.31%
More	4	100.00%	14.7	1	100.00%

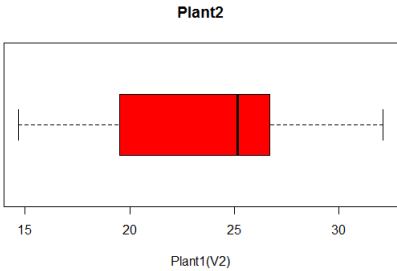
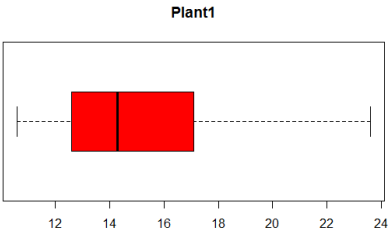


Column1	
Mean	15.35
Standard Error	1.056572834
Median	14.3
Mode	14.3
Standard Deviation	3.95333355
Sample Variance	15.62884615
Kurtosis	0.586668838
Skewness	1.069484148
Range	13
Minimum	10.6
Maximum	23.6
Sum	214.9
Count	14
Largest(1)	23.6
Smallest(1)	10.6
Confidence Level(95.0%)	2.282586834

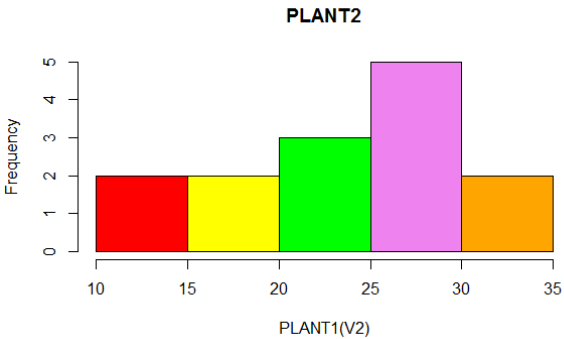
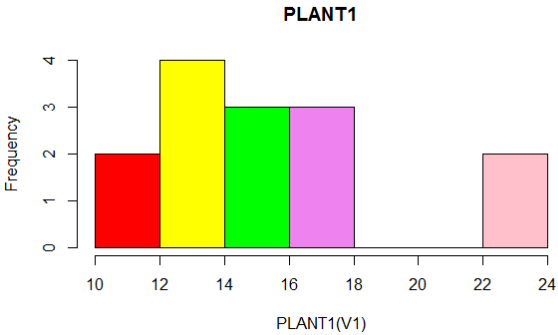
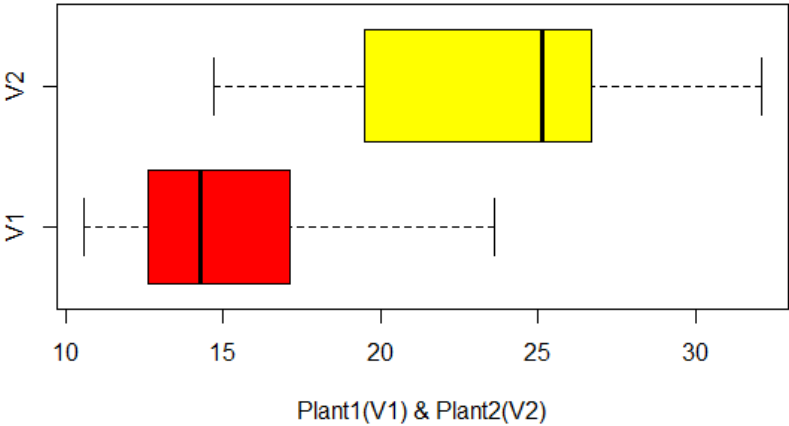
Cumulative			Cumulative		
Bin	Frequency	%	Bin	Frequency	%
0.9	1	7.69%	15.13333	7	53.85%
15.13333	7	61.54%	19.36667	3	76.92%
19.36667	3	84.62%	More	2	92.31%
More	2	100.00%	10.9	1	100.00%



Column1	
Mean	23.78571429
Standard Error	1.514285714
Median	25.15
Mode	#N/A
Standard Deviation	5.665938329
Sample Variance	32.10285714
Kurtosis	-0.72966105
Skewness	-0.42531661
Range	17.4
Minimum	14.7
Maximum	32.1
Sum	333
Count	14
Largest(1)	32.1
Smallest(1)	14.7
Confidence Level(95.0%)	3.271415394



Comparing Plants

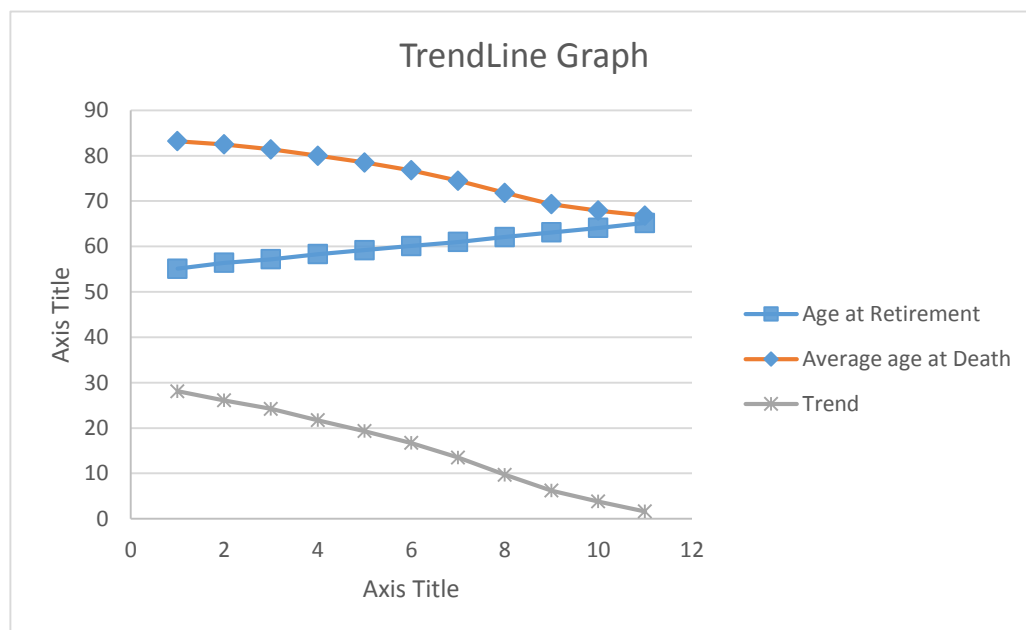


4. This data represents an actuarial study of age at retirement vs. length of life. The study was based on the number of pension checks sent to retirees of Boeing Aerospace.

Graph this data and put in a trendline. What do you notice about the graph?

Age at Retirement	Average age at Death	Trend
55.1	83.2	28.1
56.4	82.5	26.1
57.2	81.4	24.2
58.3	80	21.7
59.2	78.5	19.3
60.1	76.8	16.7
61	74.5	13.5
62.1	71.8	9.7
63.1	69.3	6.2
64.1	67.9	3.8
65.2	66.8	1.6

Based on the graph below I am able to observe that the younger you retire the longer you live, that is according to the graph/data. You can see that the trend of those that retired between 55 to 60 years old and those that retired after 60 is about 20 years of different. That is the amount of years that those who retired late lost.



5. Given this dataset, analyze snowfall for these cities. Which one gets more snow?

Season	Albany	Binghamto	Buffalo	Rochester	Syracuse
1940-41	51.7		79.3	73.7	
1941-42	45.2		89.6	66.3	
1942-43	45		85.5	70.6	
1943-44	48.6		58	46.1	
1944-45	66.8		120.7	94.7	
1945-46	54.9		110.5	49.5	
1946-47	43.1		65.4	75.5	
1947-48	90		42.1	63.4	
1948-49	44.2		40.1	50.9	
1949-50	62.6		88.7	81.7	
1950-51	48.3		71.4	75.8	
1951-52	69.4		83	75.8	100.5
1952-53	46	70.4	55.9	41.7	79.4
1953-54	48.4	65.1	89.9	77.5	91.3
1954-55	36.6	94.8	84.8	69.2	101.4
1955-56	80.2	122.6	105.2	121.4	146.8
1956-57	70.3	90.2	113.7	79.2	76.1
1957-58	74.4	111	124.7	130.8	143.8
1958-59	63.2	78.7	114.5	140.6	137.2
1959-60	60.1	105.3	115.6	161.7	134.8
1960-61	72.7	99.7	102.4	89.4	130.5
1961-62	62.6	65.6	101.4	65.6	77.3
1962-63	71.3	95.3	89.8	76.4	118.3
1963-64	77	103.3	71.5	92	83.8
1964-65	45.8	76.4	70.9	71.1	97.3
1965-66	67.1	83.7	98.3	103.2	118.8
1966-67	80.9	88.5	66.1	74	87.5
1967-68	42.2	63.6	71.6	76.7	81.2
1968-69	63.3	64.5	78.4	79.8	99.7
1969-70	87.7	115.8	120.5	119.6	127.6
1970-71	112.5	108.6	97	142.7	157.2
1971-72	89.3	106.2	109.9	105.1	133.7
1972-73	70.9	67.7	78.8	73	82.2
1973-74	58.3	86.7	88.7	99.1	123.2
1974-75	54.6	67.1	95.6	91.2	105.5
1975-76	54.2	76.3	82.5	86.2	95.8
1976-77	70.6	74.4	199.4	92.1	145
1977-78	92.4	115.3	154.3	160.9	161.2
1978-79	63.5	80	97.3	138.5	118.5
1979-80	27.4	56.8	68.4	72.2	93.4
1980-81	44.9	59.3	60.9	94.4	79
1981-82	97.1	81.6	112.4	128.4	137.3
1982-83	75	81	52.4	59.9	66
1983-84	65.2	70.9	132.5	118	113.6
1984-85	41.3	62.5	107.2	87.1	116.4
1985-86	62.5	76.3	114.7	70.7	105.3
1986-87	80.6	78.8	67.5	67.1	103.4
1987-88	76.7	81.6	56.4	69.8	113.2
1988-89	19	47.8	67.4	86.6	97.8
1989-90	57.9	74.8	93.7	105.8	153
1990-91	28.7	67	57.5	68.3	96.9
1991-92	30.7	56	92.8	110.6	167
1992-93	94.2	122.7	93.2	131.5	192.1
1993-94	88.1	131.3	112.7	126.2	163.8
1994-95	30.9	52.8	74.6	56.2	66.9
1995-96	86.5	134	141.4	130.3	170.9
1996-97	66.6	93	97.6	104.7	131.1
1997-98	52.3	92	75.6	99.7	134.7
1998-99	44.1	73.3	100.5	111.6	98.3
1999-2000	62.1	82.4	63.6	110.7	85.8
2000-01	77.1	112.6	158.7	133	191.9
2001-02	47.4	63.5	132.4	58.1	59.4
2002-03	105.4	117.6	111.3	135.2	153.2
2003-04	65.1	106.4	100.9	125.6	181.3
2004-05	75.9	106.5	109.1	113.6	136.2
2005-06	30.2	74.9	78.2	73.9	124.6
2006-07	45.9	78.6	88.9	107.2	140.2
2007-08	61.1	70.7	103.8	106	109.1
2008-09	52.6	73.3	100.2	103.7	149.6

Albany
mean 54.0513 84.951 60.8661
std. dev. 16.8072 13.7691 13.7992

Binghamton
mean 76.672 115.4078 79.4789
std. dev. 12.5609 9.0776 10.0523

Buffalo
mean 78.1535 118.2002 106.4025
std. dev. 20.2374 19.5054 26.6908

Rochester
mean 72.557 131.3117 108.5862
std. dev. 13.6591 15.2297 14.7562

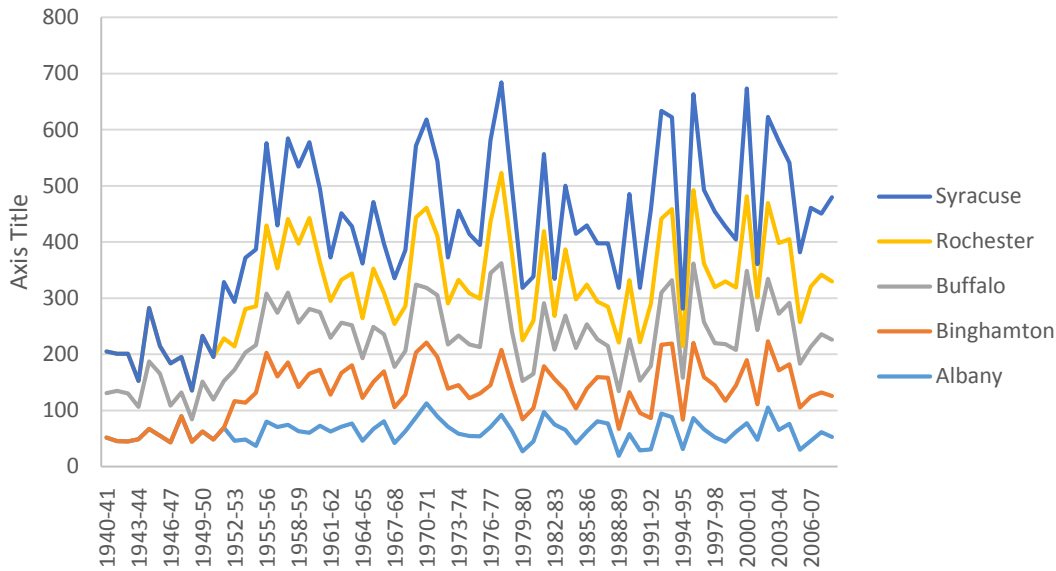
Syracuse
mean 99.4286 156.815 130.21
std. dev. 18.086 20.5699 17.5711

Clustered Instances

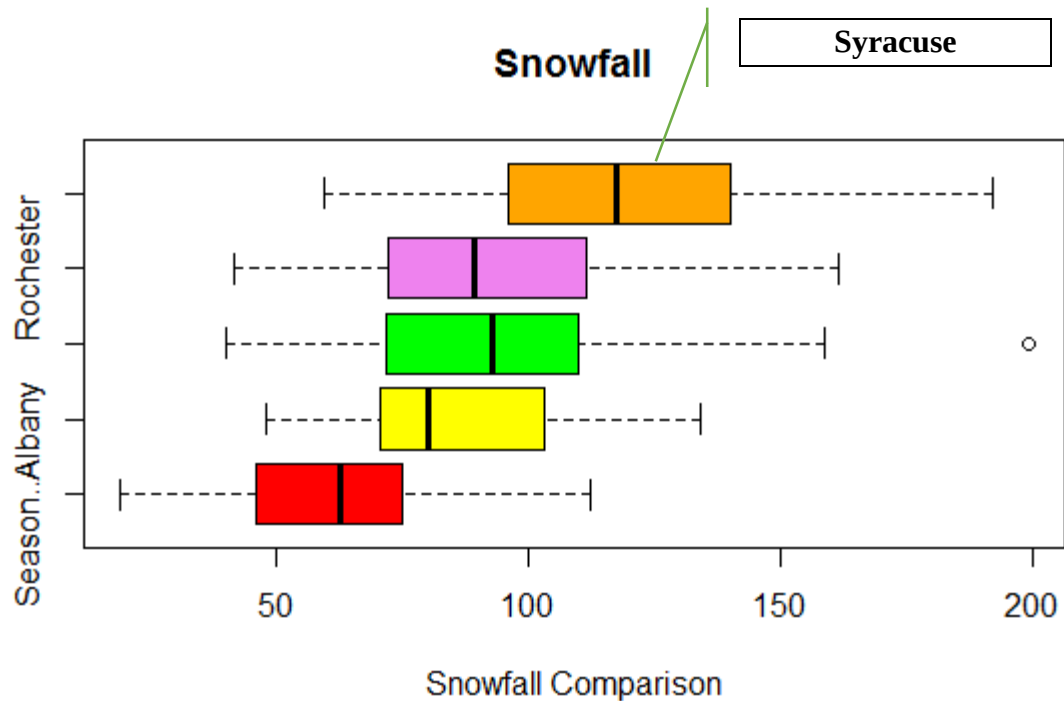
0 38 (55%)
1 14 (20%)
2 17 (25%)

As we can observe in the graphs and analysis, the city the gets snow the most is Syracuse.

Snowfall in Cities - Graph



Column1		Column2		Column3		Column4		Column5	
Mean	62.0058	Mean	85.20702	Mean	93.32609	Mean	93.46087	Mean	118.7413793
Standard Error	2.370303	Standard Error	2.792836	Standard Error	3.372589	Standard Error	3.40782	Standard Error	4.266358528
Median	62.6	Median	80	Median	92.8	Median	89.4	Median	117.35
Mode	62.6	Mode	76.3	Mode	88.7	Mode	75.8	Mode	#N/A
Standard Deviation	19.68922	Standard Deviation	21.08545	Standard Deviation	28.01483	Standard Deviation	28.30748	Standard Deviation	32.49161854
Sample Variance	387.6653	Sample Variance	444.596	Sample Variance	784.8308	Sample Variance	801.3133	Sample Variance	1055.705275
Kurtosis	-0.22093	Kurtosis	-0.57368	Kurtosis	2.15036	Kurtosis	-0.53227	Kurtosis	-0.546796567
Skewness	0.205227	Skewness	0.519153	Skewness	0.90393	Skewness	0.432303	Skewness	0.332665069
Range	93.5	Range	86.2	Range	159.3	Range	120	Range	132.7
Minimum	19	Minimum	47.8	Minimum	40.1	Minimum	41.7	Minimum	59.4
Maximum	112.5	Maximum	134	Maximum	199.4	Maximum	161.7	Maximum	192.1
Sum	4278.4	Sum	4856.8	Sum	6439.5	Sum	6448.8	Sum	6887
Count	69	Count	57	Count	69	Count	69	Count	58
Largest(1)	112.5	Largest(1)	134	Largest(1)	199.4	Largest(1)	161.7	Largest(1)	192.1
Smallest(1)	19	Smallest(1)	47.8	Smallest(1)	40.1	Smallest(1)	41.7	Smallest(1)	59.4
Confidence Level(95.0%)	4.729866	Confidence Level(95.0%)	5.594722	Confidence Level(95.0%)	6.729897	Confidence Level(95.0%)	6.800199	Confidence Level(95.0%)	8.54323559



6. An instructor misplaced his exams after correcting them. There were 8 total exams. The low grade was 50. The high was 100. The mode grade was 78 (for three students). The median for the grades was 83. He remembers that two students had grades in the 90's. What is the missing grade? Explain how you arrived at your answer.

Column1		Answer
		78
Mean	81.5	78
Standard Error	5.274602	78
Median	83	88
Mode	78	90
Standard Deviation	14.91883	90
Sample Variance	222.5714	100
Kurtosis	2.812656	
Skewness	-1.35091	
Range	50	
Minimum	50	
Maximum	100	
Sum	652	
Count	8	
Largest(1)	100	
Smallest(1)	50	
Confidence Level (95.0%)	12.47245	

The grade missing is 88, I was able to obtain this value by using the formula in excel so the value of the median is the same as the one provided (83). The value that gave me the exact result was 88.

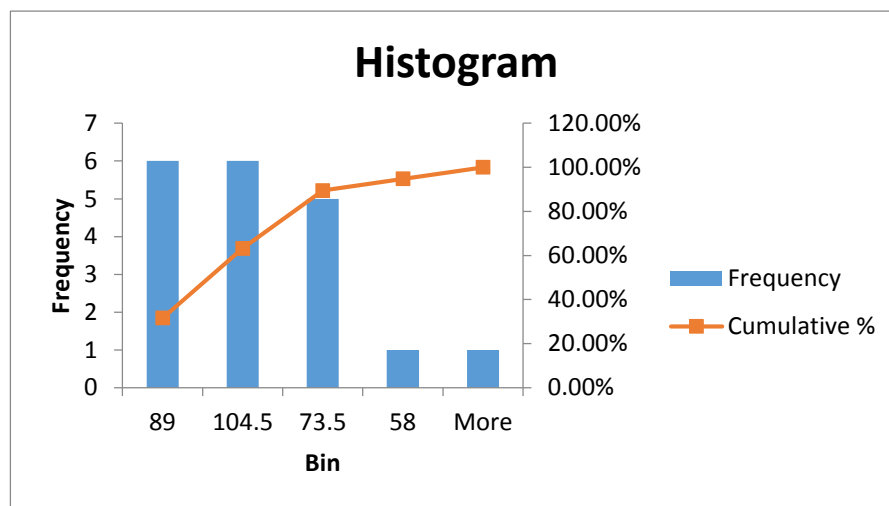
7. My home in Buffalo is approximately 72 miles away from RIT. The table below shows travel times for 20 trips back and forth.

- What is the mean travel time?
- What is the standard deviation?
- Find the 95% confidence interval for the mean

d. I think my daughter has pinkeye. The only appointment slot open at the doctor's office is for 3 p.m. I leave my office at 1:45. What is the probability that I can make it by 3 p.m.?
(Consider the travel time between my house and the doctor's office to be negligible)

Bin	Frequency	Cumulative %	Bin	Frequency	Cumulative %
58	1	5.26%	89	6	31.58%
73.5	5	31.58%	104.5	6	63.16%
89	6	63.16%	73.5	5	89.47%
104.5	6	94.74%	58	1	94.74%
More	1	100.00%	More	1	100.00%

Column1	
Mean	83.1
Standard Error	3.475402
Median	83
Mode	82
Standard Deviation	15.54247
Sample Variance	241.5684
Kurtosis	0.117622
Skewness	0.343987
Range	62
Minimum	58
Maximum	120
Sum	1662
Count	20
Largest(1)	120
Smallest(1)	58
Confidence Level(95.0%)	7.274101



In order to make you need to get there in ≤ 75 mins

Based on the graph provided, the probability for you to make it in 73.5 is of 36.58%

This means that you have more than 37% to make it in 75mins.

8. Suppose the actual amount of instant coffee that a filling machine puts into 6-ounce jars is a normally distributed random variable with a standard deviation of 0.06 ounces. If we want no more than 3% of jars to contain more than six ounces, what should be the mean fill of the jars?

The purpose is to have 3%, or less, of jars to contain no more than six ounces.

The coffee is normally distributed.

Therefore, 3% of the jars should contain 3% of the original mean.

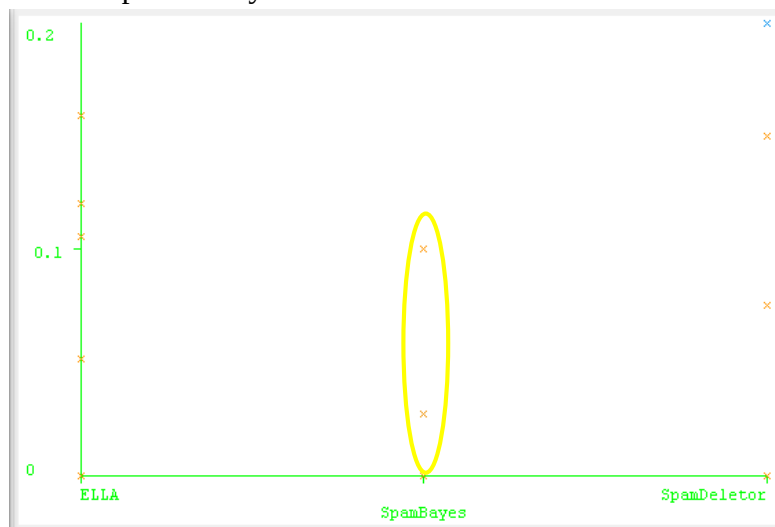
- $\text{Mean} = [1/(3\% \text{ jars})](\text{Total})$
- $3\% \text{ Mean} = (1/\text{Jars})(\text{Total})$

3% of the mean.

9. This table contains data on the effectiveness of three spam detection algorithms. Analyze the data to see which algorithm seems to work best

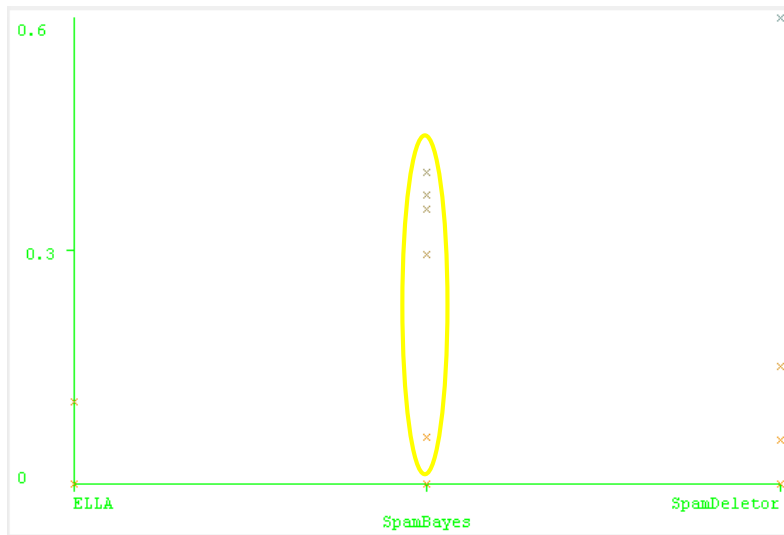
User	Spam Detector	FPR	FNR	TPR	TNR				
User1	ELLA	0	0.105263	0.894737	1			FPR	False positive results
User2	ELLA	0.05128	0	1	0.948718			FNR	False negative results
User3	ELLA	0.15909	0	1	0.840909			TPR	True positive results
User4	ELLA	0.10526	0	1	0.894737			TNR	True negative results
User5	ELLA	0	0	1	1				
User6	ELLA	0	0	1	1				
User7	ELLA	0.12	0	1	0.88				
User1	SpamBayes	0.1	0	1	0.9				
User2	SpamBayes	0.02703	0.058824	0.941176	0.972973				
User3	SpamBayes	0	0	1	1				
User4	SpamBayes	0	0.352941	0.647059	1				
User5	SpamBayes	0	0.294118	0.705882	1				
User6	SpamBayes	0	0.4	0.6	1				
User7	SpamBayes	0	0.37037	0.62963	1				
User1	SpamDeletor	0	0	1	1				
User2	SpamDeletor	0	0.055556	0.944444	1				
User3	SpamDeletor	0.075	0	1	0.925				
User4	SpamDeletor	0.15	0	1	0.85				
User5	SpamDeletor	0	0.6	0.4	0				
User6	SpamDeletor	0.2	0	0	0.8				
User7	SpamDeletor	0	0.15	0.85	1				

FPR Graph – Analysis



As we can observe in the graphs, *SpamBayes* is more effective based on the results. It has less false positive response than the others, more false “negative” response, more “TRUE” POSITIVE response, and less true negative response.

FNR Graph – Analysis



TPR Graph – Analysis



TNR Graph - Analysis

